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GOOGLE SCHOLAR: <https://scholar.google.com/citations?user=HDQU65AAAAAJ>

Research Interests

My research focuses on **Human-Computer Interaction**, particularly on designing (1) **body-centric interfaces** to **augment human** capabilities in **mixed reality** (MR) through **embodied interaction**; (2) leveraging **AI-driven context awareness** to enable **adaptive user interfaces** in **ubiquitous MR**, facilitating seamless **situated interaction** in *everyday* environments (aka. **spatial computing**).

Education

- 2022.10 -
2026.03 **Ph.D.** in Engineering, *Robinson College*, **University of Cambridge**
Department of Engineering (Division: [Information Engineering](#))
- 2018.9 -
2022.7 **B.Sc. (Hons.)** in Information and Computing Science, **University of Liverpool**
B.Sc. (Hons.) in Information and Computing Science, **Xi'an Jiaotong-Liverpool University**

Research Experience

- 2025.12 -
2026.3 **Incoming Research Scientist Intern**, [Kyoto Research](#), [Sony CSL](#)
Mentor: [Jun Rekimoto](#)
- 2022.10 -
Present **Ph.D. Researcher**, [Intelligent Interactive Systems Group](#), University of Cambridge
Supervisor: [Per Ola Kristensson](#)
Advisor: [John J. Dudley](#)
- 2022.10 -
Present **Student Fellow**, [The Leverhulme Centre for the Future of Intelligence](#)
Host: [Dorian Peters](#)
- 2022.7 -
2022.9 **Visiting Ph.D. Researcher**, Hong Kong University of Science and Technology
Host: [Pan Hui](#)
- 2021.4 -
2021.12 **Research Intern**, [DIVA HCI Group](#), Institut Polytechnique de Paris
Research Intern, [Augmented Perception Lab](#), Carnegie Mellon University
Co-Advisors: [Jan Gugenheimer](#) (now at TU Darmstadt) & [David Lindlbauer](#)

2020.4 -
2020.12

Research Intern, [Exertion Games Lab](#), Monash University
Advisor: [Florian ‘Floyd’ Mueller](#)

2019.4 -
2022.7

Research Assistant, X-CHI Lab, Xi’an Jiaotong-Liverpool University
Advisor: [Hai-Ning Liang](#) (now at HKUST(GZ))

Publications

I have published/submitted the majority of my work in top-tier conferences and journals in VR/AR/MR (e.g., **IEEE TVCG**, **VR**, **ISMAR**, **ACM SUI**) and HCI (e.g., **ACM TOCHI**, **CHI**, **ISS**, **CHI PLAY**)¹ These venues typically have acceptance rates around 25%. For the latest publications, please visit my [\[Google Scholar\]](#)² or [\[DBLP\]](#) profiles.

FULL PAPERS

- P.18 **Li, X.**, He, W., Kristensson, P. O. (2025), “Evaluating the Usability of Microgestures for Text Editing Tasks in Virtual Reality”, *IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG)*. *To appear*. **[2025 UKRI Innovate UK Immersive Tech Award: Technical Innovation]**
- P.17 **Li, X.**, Kristensson, P. O. (2025), “Bend It, Aim It, Tap It: Designing an On-Body Disambiguation Mechanism for Curve Selection in Mixed Reality”, In *Proceedings of the ACM Symposium on Spatial User Interaction (ACM SUI’25)*. **[Selected as the University of Cambridge’s delegate at SIYSS 2025, during Nobel Week, Stockholm.]**
- P.16 Wang, J.-D., Zhou, K., Ren, H., Kristensson, P. O., **Li, X.*** (2025), “Handows: A Palm-Based Interactive Multi-Window Management System in Virtual Reality”, *IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG)*, with *IEEE International Symposium on Mixed and Augmented Reality (IEEE ISMAR’25)*. (*: Corresponding Author)
- P.15 Deng, D., Yang, X., **Li, X.**, Li, M., Huang, Y., Liu, L., Bujić, M., Hamari, J. (2025). “How VR Boxing Exergames Meet (or Miss) Psychological Needs: A Text Mining Analysis of User Reviews”, In *International Conference on Extended Reality (CCF ICXR’25)*.
- P.14 Wang, H., Du, J., Li, Y., Zhang, L., **Li, X.*** (2025). “Grand Challenges in Immersive Technologies for Cultural Heritage.” *International Journal of Human-Computer Interaction (IJHCI)*. (*: Corresponding Author)
- P.13 **Li, X.**, He, W., Jin, S., Gugenheimer, J., Hui, P., Liang, H.-N., Kristensson, P. O. (2024). “Investigating Creation Perspectives and Icon Placement Preferences for On-Body Menus in Virtual Reality.” In *Proceedings of the ACM on Human-Computer Interaction (PACMHCI)*, with *ACM International Conference on Interactive Surfaces and Spaces (ACM ISS’24)*.
- P.12 **Li, X.**, Wang, J.-D., Dudley, J. J., Kristensson, P. O. (2024). “Swarm Manipulation: An Efficient and Accurate Technique for Multi-Object Manipulation in Virtual Reality.” **Computers & Graphics**.

¹Students I supervised are underlined.

²440 citations, h-index = 11, i10-index = 13, according to Google Scholar, September 17, 2025.

- P.11 He, W., **Li, X.**, Xu, S., Chen, Y., Sio, C.-I., Kan, G. L., Lee, L.-H. (2024). “MetaDragonBoat: Exploring Paddling Techniques of Virtual Dragon Boating in a Metaverse Campus.” In *Proceedings of the ACM International Conference on Multimedia (ACM MM’24)*.
- P.10 **Li, X.**, Wang, J.-D., Dudley, J. J., Kristensson, P. O. (2023). “Swarm Manipulation in Virtual Reality.” In *Proceedings of the ACM Symposium on Spatial User Interaction (ACM SUI’23)*.
- P.9 Patibanda, R., Memories, M., Hill, C., Saini, A., **Li, X.**, Chen, Y., Matvienko, A., Knibbe, J., van den Hoven, E., Mueller, F. F. (2023). “Auto-Paizo Games: Towards Understanding the Design of Games that Aim to Unify a Player’s Physical Body and the Virtual World.” In *Proceedings of the ACM on Human-Computer Interaction (PACMHCI)*, with *ACM Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY’23)*. **[Best Interactivity Audience Award]**
- P.8 Patibanda, R., Saini, A., Overdevest, N., Montoya, M., **Li, X.**, Chen, Y., Nisal, S., Andres, J., Knibbe, J., van den Hoven, E., Mueller, F. F. (2023). “Fused Spectatorship: Designing Active Bodily Spectating Experiences.” In *Proceedings of the ACM on Human-Computer Interaction (PACMHCI)*, with *ACM Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY’23)*.
- P.7 Mueller, F. F., Semertzidis, N., Andres, J., Marshall, J., Benford, S., **Li, X.**, Mehta, Y., Matjeka, L. (2023). “Towards Understanding the Design of Intertwined Human-Computer Integration.” *ACM Transactions on Computer-Human Interaction (ACM TOCHI)*.
- P.6 Gao, Z.* **Li, X.***, Liu, C., Wang, X., Wang, A., Yang, L., Wang, Y., Hui, P., Braud, T. (2023). “VR PreM+: An Immersive Pre-Learning Branching Visualization System for Museum Tours.” In *Proceedings of the ACM International Conference on Human-Computer Interaction with Chinese Speakers (ACM Chinese CHI’23)*. (*: Equal Contribution)
- P.5 Dray, S., Buie, E., Crampton Smith, G., Kroma, A., **Li, X.**, Marcus, A., Pereira, M., Rivera-Loaiza, C., Rohn, J. (2022). “‘It Changed My Life’: Stories of the Influence of SIGCHI.” **ACM Interactions**.
- P.4 Mueller, F. F., Patibanda, R., Byrne, R., Li, Z., Wang, Y., Andres, J., **Li, X.**, Marquez, J., Greuter, S., Duckworth, J., Marshall, J. (2021). “Limited Control Over the Body as an Intriguing Play Design Resource.” In *Proceedings of the ACM Conference on Human Factors in Computing Systems (ACM CHI’21)*.
- P.3 Lu, X., Yu, D., Liang, H.-N., Xu, W., Chen, Y., **Li, X.**, Hasan, K. (2020). “Exploration of Hands-Free Text Entry Techniques for Virtual Reality.” In *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality (IEEE ISMAR’20)*.
- P.2 Xu, W., Liang, H.-N., He, Q., **Li, X.**, Chen, Y., Yu, K. (2020). “Results and Guidelines From a Repeated-Measures Design Experiment Comparing Standing and Seated Full-Body Gesture-Based Immersive Virtual Reality Exergames: Within-Subjects Evaluation.” **JMIR Serious Games**.
- P.1 Xu, W., Liang, H.-N., Chen, Y., **Li, X.**, Yu, K. (2020). “Exploring Visual Techniques for Boundary Awareness During Interaction in Augmented Reality Head-Mounted Displays.” In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR’20)*. **[Best Conference**

Paper Nomination]

EXTENDED ABSTRACTS (PEER-REVIEWED)

- EA.12 **Li, X.**, Kristensson, P. O. (2025). “Optimizing Curve-Based Selection with On-Body Surfaces in Virtual Environments.” In *Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems (ACM CHI’25)*.
- EA.11 Patibanda, R., Saini, A., **Li, X.**, Chen, Y., van den Hoven, E., Mueller, F. F. (2025). “Playing with Yourself: How Electrical Muscle Stimulation Can Turn Your Body Into A Playful Interface.” In *Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems (ACM CHI’25)*.
- EA.9 **Li, X.** (2025). “[DC] Beyond Avatars: Designing Bodily Interfaces in Virtual Environments.” In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (IEEE VR’25 Doctoral Consortium)*.
- EA.8 **Li, X.** (2024). “Swarm Bodily Interfaces in Virtual Environments.” Bianchi-Berthouze, N., Gonzalez-Franco, M., Mueller, F. F., Sra, M. (Eds.). **Dagstuhl Seminar Report**.
- EA.7 Lin, Z., He, W., Liu, X., Ye, M., **Li, X.**, Kan, G. L. (2024). “The Framework of NAVIS: Navigating Virtual Spaces with Immersive Scooters.” In *Proceedings of the 2024 International Conference on Mobile and Ubiquitous Multimedia (ACM MUM’24)*.
- EA.6 **Li, X.**, Chen, Y., Tang, X. (2023). “GesMessages: Using Mid-Air Gestures to Manage Notifications.” In *Proceedings of the 2023 Symposium on Spatial User Interaction (ACM SUI’23)*.
- EA.5 **Li, X.**, Chen, Y., Tang, X. (2022). “GesPlayer: Using Augmented Gestures to Empower Video Players.” In *Companion Proceedings of the 2022 International Conference on Interactive Surfaces and Spaces (ACM ISS’22)*.
- EA.4 **Li, X.**, Tang, X., Tong, X., Patibanda, R., Mueller, F. F. (2021). “Myopic Bike and Say Hi: Games for Empathizing with the Myopic.” *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY’21)*. **[Student Game Design Competition Award Finalist]**
- EA.3 Patibanda, R., **Li, X.**, Chen, Y., Saini, A., Hill, C., van den Hoven, E., Mueller, F. F. (2021). “Actuating Myself: Designing Hand-Games Incorporating Electrical Muscle Stimulation.” In *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY’21)*.
- EA.2 **Li, X.**, Chen, Y., Patibanda, R., Mueller, F. F. (2021). “vrCAPTCHA: Exploring CAPTCHA Designs in Virtual Reality.” In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (ACM CHI’21)*.
- EA.1 Xu, W., Liang, H.-N., Ma, X., **Li, X.** (2020). “VirusBoxing: A HIIT-Based VR Boxing Game.” In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY’20)*. **[Student Game Design Competition Award Finalist]**

UNDER REVIEW

The following manuscripts are currently under double-blind review and are listed here for completeness. Non-peer-reviewed preprints are available upon request; please treat this information as confidential, in line with the principle of anonymity and the need to avoid conflicts of interest.

- One **first-authored** scoping review on evaluating immersive experiences.
- Two **co-first-authored** papers on (1) collaborative VR exergames and (2) locomotion systems.
- One **corresponding-authored** paper on virtual retail agents for metaverse shopping.
- One co-authored paper on grand challenges in human-computer bodily control.
- One co-authored journal article on the future of SportsHCI.
- One co-authored journal article on designing bodily intelligence.

Achievements

PAPER AWARDS

2023	Best Interactivity Audience Award,	<i>ACM CHI PLAY'23</i>
2021	Student Game Design Competition Award Finalist,	<i>ACM CHI PLAY'21</i>
2020	Student Game Design Competition Award Finalist,	<i>ACM CHI PLAY'20</i>
2020	Best Conference Paper Nomination,	<i>IEEE VR'20</i>

HONOURS

2025	SIYSS ³ Young Scientist (during Nobel Week),	<i>Swedish Federation for Young Scientists</i>
2025	Innovate UK Immersive Tech Award: Technical Innovation,	<i>UKRI Innovate UK</i>
2024	Fatima Fellowship Mentor,	<i>Fatima Fellowship Program</i>
2023	Student Volunteer Award,	<i>ACM CHI'23</i>
2023	Cambridge-TECH Scholar (<i>Declined</i>),	<i>University of Cambridge & Tokyo Tech</i>
2022	Student Fellowship,	<i>Leverhulme Centre for the Future of Intelligence</i>
2021	Student Volunteer Award,	<i>ACM CHI'21</i>
2021	Student Representative in Academic Practice Committee,	<i>Xi'an Jiaotong-Liverpool University</i>

GRANTS, FUNDS, & SCHOLARSHIPS

2025	IET International Travel Award (£1500) (only 2 for each round),	<i>IET</i>
2025	Cambridge Trust Conference Fund (£500),	<i>Cambridge Trust, University of Cambridge</i>
2025	Conference Fund Award (£1000),	<i>Department of Engineering, University of Cambridge</i>
2025	Academic Expenses Fund (£500),	<i>Robinson College, University of Cambridge</i>
2024	Symons Fund (£470),	<i>Robinson College, University of Cambridge</i>
2023	Symons Fund (£200),	<i>Robinson College, University of Cambridge</i>
2022	Cambridge Trust International Scholarship (with UKRI Stipend),	<i>Cambridge Trust & CSC</i>
2022	Gary Marsden Travel Award (\$3500),	<i>ACM SIGCHI</i>
2021	I.D.E.A. Scholarship,	<i>IEEE ISMAR'21</i>
2021	ACM Registration Scholarship,	<i>ACM UIST'21</i>

³Stockholm International Youth Science Seminar: <https://ungaforskare.se/siyss/>

2021	Ableton Scholarship,	<i>NIME'21</i>
2021	Bridge to VR Scholarship,	<i>IEEE VR'21</i>
2020	Research Assistant Fellowship (¥12000),	<i>Xi'an Jiaotong-Liverpool University</i>
2019	Summer Undergraduate Research Fellowship (¥8000),	<i>Xi'an Jiaotong-Liverpool University</i>

Teaching

TEACHING SUPERVISION⁴

2025 - Present	Day-to-Day Supervisor , Queens' College, University of Cambridge <i>Course</i> : Part II Projects (Final Year Dissertation), Principal Supervisor: Cengiz Oztireli
2026 Easter	Supervisor , Trinity College, University of Cambridge <i>Course</i> : IA: Interaction Design , Principal Lecturer: Hatice Gunes
2026 Lent	Supervisor , Trinity College, University of Cambridge <i>Course</i> : IB: Further Human-Computer Interaction , Principal Lecturer: Alan Blackwell
2024 Easter	Supervisor , Trinity College, University of Cambridge <i>Course</i> : IA: Interaction Design , Principal Lecturer: Hatice Gunes
2024 Lent	Supervisor , Trinity College, University of Cambridge <i>Course</i> : IB: Further Human-Computer Interaction , Principal Lecturer: Alan Blackwell

RESEARCH SUPERVISIONS

2025 - Present	Taylor Hope,	Now: Undergraduate at the <i>University of Cambridge</i>
2024 - Present	Wei He ,	Next: Ph.D. at the <i>Hong Kong University of Science and Technology</i>
2023 - Present	Shan Jin,	Next: Ph.D. at the <i>University of Cambridge</i>
2023 - 2025	Hanbing Wang,	Next: Ph.D. at <i>Tsinghua University</i>
2024	Fatima Fellowship Research Projects ,	<i>Fatima Fellowship Program</i>
2020 - 2024	Yuzheng Chen ,	Next: M.Res. at <i>Lancaster University</i>
2023 - 2025	Jin-Du Wang ,	Next: M.Phil. at the <i>Hong Kong University of Science and Technology</i>
2020 - 2022	Xiaohang Tang ,	Next: Ph.D. at <i>Virginia Tech</i>

Academic Services

CONFERENCE ORGANIZING ROLES

2026	Assistant to Technical Program Chairs,	<i>ACM CHI'26</i>
2024	Social Media Co-Chair,	<i>ACM CHI PLAY'24</i>
2024	Social Communications Co-Chair,	<i>ACM CHI'24</i>
2023	Social Media Co-Chair,	<i>ACM CHI PLAY'23</i>
2023	“Pick-A-Brain” Organizer,	<i>ACM CHI'23</i>
2022	Online Participation Co-Chair,	<i>ACM DIS'22</i>
2022	“Ask-Me-Anything” (AMA) Organizer,	<i>ACM DIS'22</i>

⁴At Cambridge, supervisions are independently designed and led small-group tutorials (bi-weekly), distinct from standard TA roles. See [this article](#) for details.

TECHNICAL PROGRAM COMMITTEES & ASSOCIATE CHAIR ROLES

2025	Full Paper,	<i>IEEE ISEMV'25 (with ICCV'25)</i>
2025	Full Paper,	<i>ACM IMX'25</i>
2025	Poster and Demo,	<i>ACM UbiComp/ISWC'25</i>
2022 - 2025	Late-Breaking Work,	<i>ACM CHI'22, '23, '24, '25</i>
2024	Work-in-Progress,	<i>ACM CHI PLAY'24</i>
2023	Poster and Demo,	<i>ACM UbiComp/ISWC'23</i>
2021	Poster and Demo,	<i>ACM IUI'21</i>

SESSION CHAIR ROLES

2025	Working with AI (or not),	<i>ACM CHI'25</i>
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PROFESSIONAL MEMBERSHIPS

ACM SIGCHI, IEEE, BCS, IET

REVIEWING ACTIVITIES

I have reviewed **180+** papers and received **10x** Highly Useful or Special Recognitions for Outstanding Reviews (*).

Full Papers

2025	CHI, UIST, TVCG, IEEE VR, ISMAR, MM, MobileHCI, DIS*, IMX, Virtual Reality, ISEMV, ISS
2024	CHI, UIST*, CSCW*, MM, DIS, ISS, CHI PLAY, SUI, EICS
2023	CHI, UIST, TVCG, ISMAR, VRST
2022	CHI, UIST*, TVCG, IEEE VR, ISMAR, ISS, CSCW, SUI, VRST
2021	CHI, TVCG*, IEEE VR, ISMAR*, CHI PLAY, VRST
2020	TVCG*, ISMAR*, IMWUT, VRST

Extended Abstracts & Short Papers

2025	CHI, UbiComp
2024	CHI*, UbiComp, CHI PLAY*, MuC
2023	CHI, UbiComp, CHI PLAY, Chinese CHI
2022	CHI, UbiComp, CHI PLAY, ISMAR
2021	CHI, IUI
2020	CHI PLAY

STUDENT VOLUNTEER

2025	IEEE VR, ISMAR
2023	CHI
2022	CHI, UIST
2021	CHI, CHI PLAY, ISMAR, MobileHCI, DIS, TEI
2020	CHI PLAY, ISMAR, IEEE AIxVR

Presentations

INVITED TALKS

- 2024-2025 **Beyond avatars: Designing bodily interfaces in virtual environments.**
The Hong Kong University of Science and Technology (Guangzhou)
Goldsmiths, University of London
Northeastern University, China
IEEE VR 2025 Doctoral Consortium
The Chinese University of Hong Kong, Shenzhen
University of Michigan-Shanghai Jiao Tong University Joint Institute
Nanjing University of Science and Technology
- Bend It, Aim It, Tap It: Designing on-body interactions in virtual reality.**
Stockholm International Youth Science Seminar, Nobel Week
- microGEXT: Using microgestures for text editing in VR.**
Innovate UK Immersive Tech
- 2024 **Swarm bodily interfaces in virtual environments.**
Dagstuhl Seminar
- 2022 **Towards understanding body-centric systems.**
The Hong Kong University of Science and Technology (Guangzhou)
- 2021 Introduction to human-computer integration.
Xi'an Jiaotong-Liverpool University, Museum
- 2020 An undergraduate student's academic path.
Xi'an Jiaotong-Liverpool University, Library

PAPERS & DEMO PRESENTATIONS

- 2025 IEEE VR, ACM CHI, ACM SUI, IEEE ISMAR, [Innovate UK](#), [CHIA Conference](#)
- 2024 ACM ISS, [Cambridge Festival](#), [CHIA Conference](#)
- 2023 ACM SUI, CUED Div C Conference
- 2022 ACM ISS
- 2021 ACM CHI, ACM CHI PLAY
- 2020 IEEE VR, ChinaVR

Media Coverages

- 2025 [Winners celebrated at the buzzing Innovate UK Immersive Tech Awards 2025.](#) *Innovate UK*
- 2025 [Robinson student awarded the Innovate UK Immersive Tech Award.](#) *Robinson College News*
- 2025 [Finalists announced for the Innovate UK Immersive Tech Awards 2025.](#) *Innovate UK (UKRI)*
- 2025 [Robinson PhD student finalist in Technical Innovation.](#) *Robinson College News*
- 2023 [Cutting mind-body control to unlock a new level of immersion.](#) *Monash University*
- 2022 ["It changed my life:" Stories of the Influence of SIGCHI.](#) *ACM Interactions Magazine*
- 2021 [Introduction to Human-Computer Integration \(in Chinese\)](#) *XJTLU Museum*
- 2021 [Selected IEEE VR papers preached by Chinese scholars \(in Chinese\)](#) *CCF-TCVRV Secretariat*
- 2020 [Living Library Highlights \(in Chinese\)](#) *XJTLU Library*

References

- R.1 **Per Ola Kristensson**, University of Cambridge
Co-Director of Centre for Human-Inspired Artificial Intelligence & Trinity College Fellow
Professor of Interactive Systems Engineering
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- R.2 **Florian ‘Floyd’ Mueller**, Monash University
ACM SIGCHI Academy & Director of Research in Human-Centred Computing
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- R.3 **Hai-Ning Liang**, The Hong Kong University of Science and Technology (Guangzhou)
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- R.4 **John J. Dudley**, University of Cambridge
Course Director for the MPhil in Machine Learning and Machine Intelligence
Associate Teaching Professor
Email: jjd50@cam.ac.uk